

Product Carbon Footprint Study Report

Product Name: integrated circuit production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: integrated circuit production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg integrated circuit production, memory type

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg integrated circuit production, memory type.

2. System Boundary

integrated circuit production, memory type, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of integrated circuit production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1108.968 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

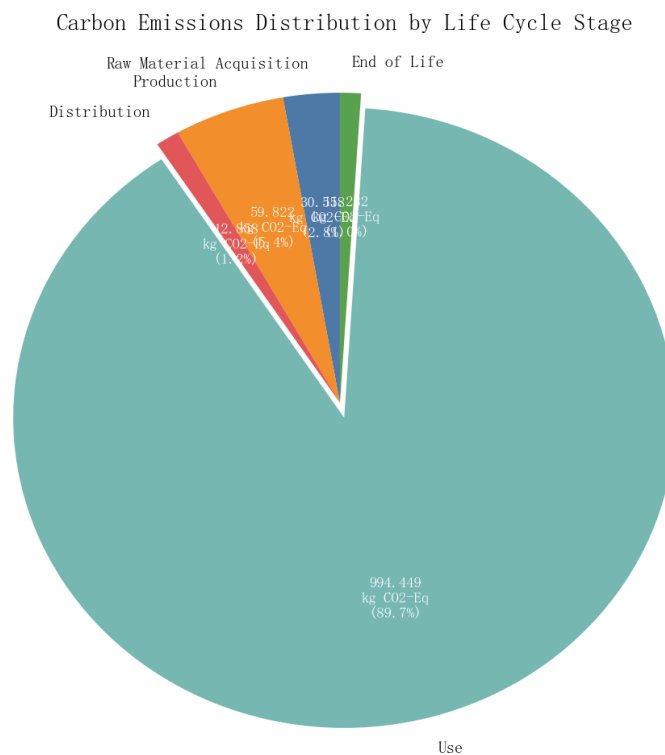


Table 1: Carbon Emissions by Life Cycle Stage

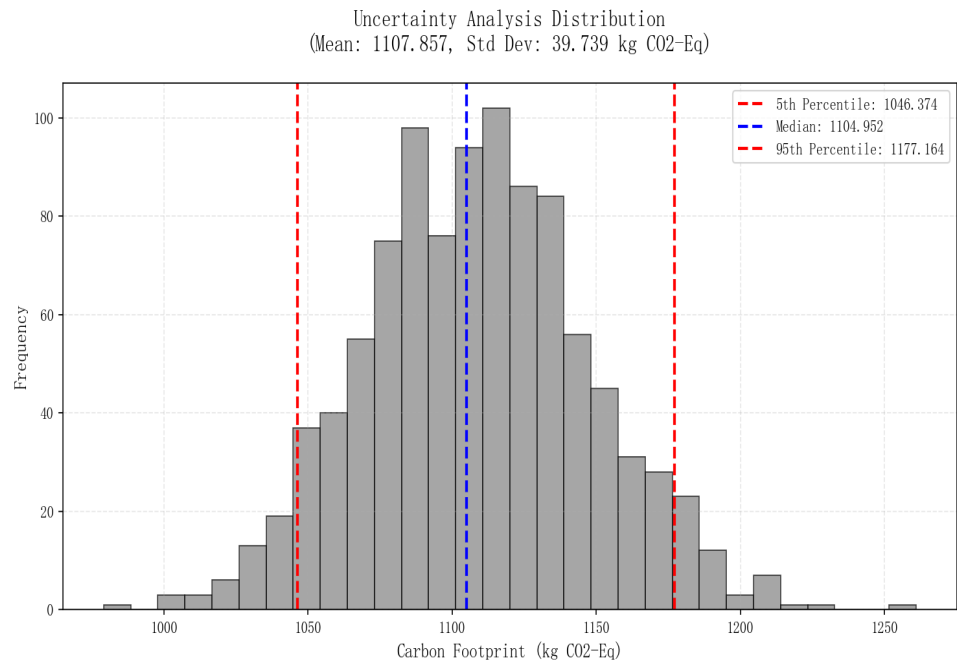
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	30.558	2.76
Production	59.822	5.39
Distribution	12.858	1.16

Use	994.449	89.67
End of Life	11.282	1.02
Total	1108.968	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	1107.857
Standard Deviation	39.739
Coefficient of Variation (CV)	3.59%
Median	1104.952
5th Percentile	1046.374
95th Percentile	1177.164

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the integrated circuit production is 1108.968 kg CO₂-Eq, representing a substantial environmental impact primarily driven by operational energy demands over the product lifecycle. This level indicates significant greenhouse gas emissions associated with semiconductor manufacturing and usage, highlighting the need for targeted mitigation in high-impact electronics sectors.

Lifecycle contributions reveal a pronounced dominance of the use phase at 994.449 kg CO₂-Eq (89.7%), while raw material extraction (2.8%), production (5.4%), distribution (1.2%), and end-of-life (1.0%) collectively account for just 10.3%. This distribution underscores that indirect emissions from electricity consumption during device operation far exceed direct manufacturing impacts.

Major emission sources are concentrated in the use stage, likely stemming from power-intensive applications of integrated circuits in consumer electronics and computing systems, with secondary contributions from energy-intensive wafer fabrication and material processing.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a standard 5-10 year product lifetime for integrated circuits, average global electricity grid mixes for the use phase, and representative transportation distances for distribution. These parameters enable consistent modeling but rely on generalized scenarios rather than site-specific operational data.

Limitations of data sources and quality stem from reliance on aggregated industry databases such as Ecoinvent, which may not fully capture proprietary manufacturing processes or regional variations in semiconductor fabrication. Data gaps in emerging low-carbon materials and end-of-life recycling efficiencies further constrain accuracy, potentially underestimating or overestimating certain flows.

System boundary restrictions encompass a cradle-to-grave approach limited to the five defined stages, excluding upstream infrastructure, indirect land-use changes, and downstream recycling credits beyond basic disposal. This truncation omits potential rebound effects from product reuse and broader supply chain interactions.

3. Suggestions for Improvement

- Optimize use-phase energy efficiency through design of low-power integrated circuits and integration of renewable energy sourcing for end-user devices to address the 89.7% emission share.
- Implement advanced process optimizations in wafer fabrication, such as renewable-powered cleanrooms and waste heat recovery, to reduce the production stage's 5.4% contribution.
- Enhance raw material traceability by sourcing low-carbon alternatives like recycled silicon and adopting circular economy strategies for end-of-life recovery to minimize the 2.8% material impact.

- Improve data quality by incorporating primary site-specific measurements and region-specific electricity factors into future LCAs, reducing reliance on generic databases.
- Explore lightweight packaging and localized manufacturing hubs to lower distribution emissions while piloting extended producer responsibility programs for better end-of-life management.